Innovation on Tools and the Rise of Skill Premium

Hyejin Park and Younghun Shim

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Motivation and Research Question

- Rising skill premium since 1980 (40%→85%) graph
- Main hypothesis in the literature: skill-biased technical change
  - Technology ↑ + (more) complement with skilled workers
    → productivity of skilled worker ↑ → skill premium ↑

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- **This paper:**

  Using text analysis, measure innovation on tools at occupation level
- **Question:**
  How much tool innovation contributes to rising skill premium?
  - Is it more directed toward skill-intensive jobs?
  - Is it more complementary to skilled workers?

Structure

- ★ **Data**
- ★ **Text Matching Procedure**
- ★ **Reduced Form Evidence**
- ★ **Structural Framework** (Skipped)
- ★ **Results** (Skipped)

Data: Tools

- Tool - occupation data from O*NET
  - Objects: *Specific tools or technology essential to occupational incumbents for performing their work roles*
  - *"Tools" are generally comprised of machines, equipment, and tools*
  - 775 occupations, each have 39 tools on average
  - 4,180 unique tools in total

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Tool name	# of Jobs using
Personal computers	565
Scanners	208
Screwdrivers	180
Power drills	139
Wire scissors	1
Folding machines	1
Solvent recyclers	1
Dental plaster knives	1

  - 83% of tools are used by less than 10 occupations

Data: Innovation

- United States Patent and Trademark Office (USPTO)
  - Universe data of patent registered in the U.S.
  - Title, abstract, technology field
  - 6.1M utility patents in 1970-2015

Data: Wage, College Premium, Employment

- Decennial Census 1980, American Community Survey 2015-2019
- Employment and average wage by occupation, year, industry, and education group (College+ vs. Below)

Table: Summary Statistics of changes in 1980-2015

	Mean	Sd	Median	1Q.	3Q.	N.
Employment Growth	0.77	8.38	0.46	-0.13	1.13	324
Wage Growth	2.71	1.03	2.55	2.11	3.22	324
D. College Premium (ppt)	9.50	14.17	8.99	2.96	12.69	323
D. Skill Intensity (ppt)	22.10	6.12	22.23	19.47	26.50	324

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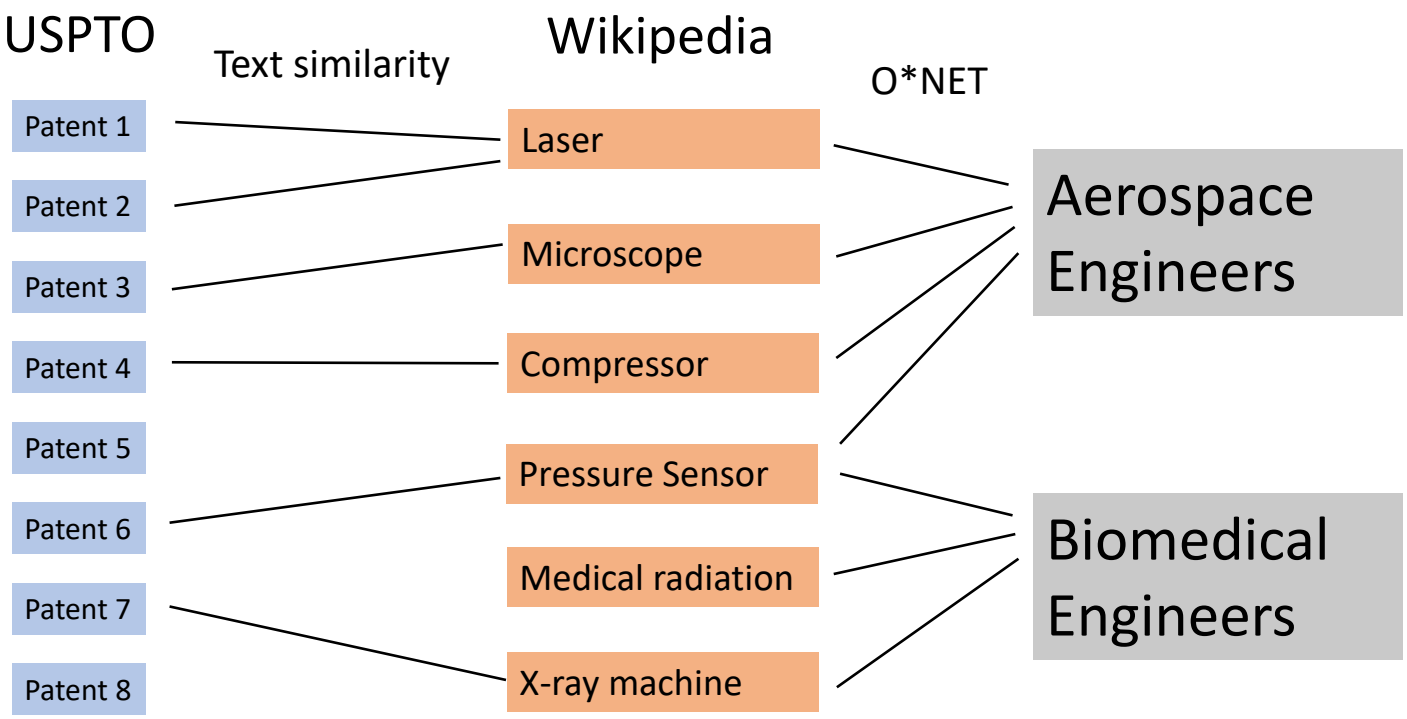
Summary of Matching



Example of Matching

Patent: System and method for detecting deterioration of oxygen sensor

feedback type air-fuel ratio control system control air-fuel ratio air-fuel mixture feed internal combustion engine accordance information signal issue first oxygen sensor install exhaust line engine exhaust line catalytic converter position downstream first oxygen sensor provide system control system detect deterioration first oxygen sensor

Wikipedia: Oxygen sensor

oxygen sensor lambda sensor lambda refer air-fuel equivalence ratio usually denote electronic device measure proportion oxygen gas liquid analyse common application measure exhaust gas concentration oxygen internal combustion engine automobile vehicle order calculate required dynamically adjust air-fuel ratio catalytic converter work optimally.

Similarity: 0.25. Oxygen sensor is used by biochemical engineers

Summary Statistics of Matching Result

- Wikipedia page is found for 1,825 / 4,180 tools
- 1.4M patents (22.8%) are matched with at least one tool
- Use average number of patents at occupation level example

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Table: The number of patent matched to each tool

	Mean	Sd	Median	1Q.	3Q.	N.	Matching rate (%)
Patent (1970s)	39.53	94.94	7.92	2.00	30.65	1,802	23.83%
Patent (1980s)	81.93	190.84	17.18	4.23	66.00	1,802	23.87%
Patent (1990s)	152.86	410.81	30.70	8.67	115.23	1,802	23.49%
Patent (2000s)	264.11	806.38	43.90	13.67	175.75	1,802	23.00%

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Table: The number of patent matched to each occupation

	Mean	Sd	Median	1Q.	3Q.	N.
Patent (1970s)	28.95	26.00	21.83	13.04	36.31	776
Patent (1980s)	68.65	49.46	55.34	34.85	85.30	776
Patent (1990s)	200.66	124.25	178.55	106.69	266.95	776
Patent (2000s)	324.01	258.21	260.34	146.63	410.34	776

histogram

Top and bottom occupations with patent growth

- Top 5 occupation : First-Line Supervisor, Computer and Information Systems Manager, Bill and Account Collector, Marketing Specialist, Sales Manager
- Bottom 5 occupation : Sewing Machine Operator, Dishwasher, Furnace operator, Tire Builder, Upholsterer

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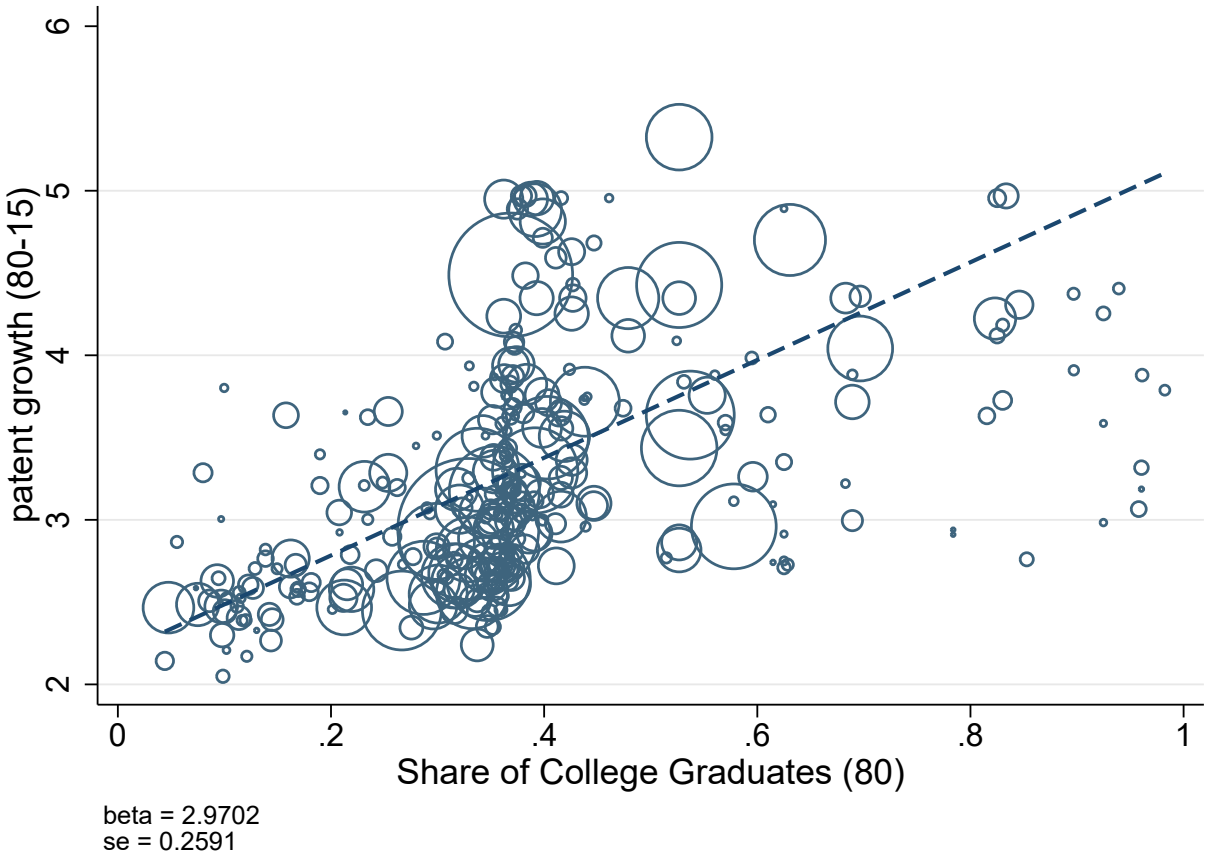
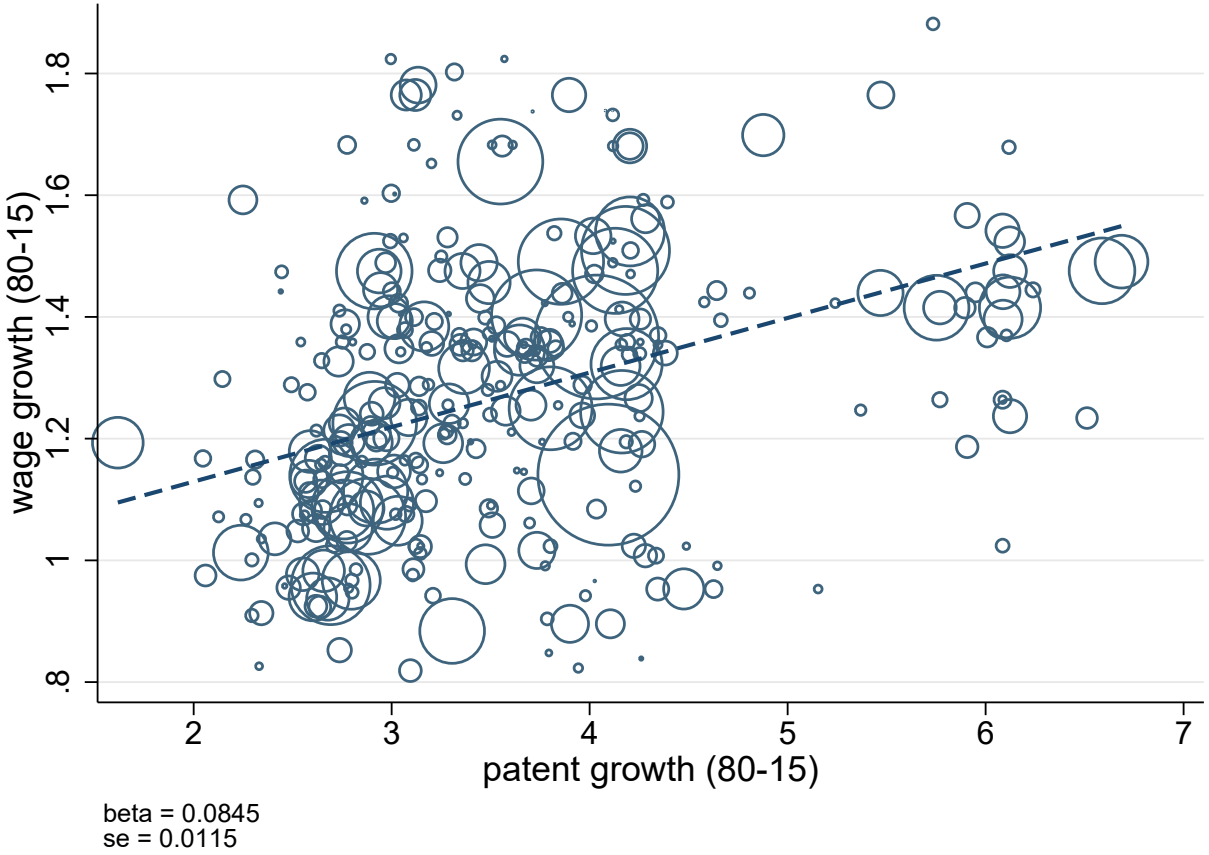
Patent Grows More in Skill-Intensive Jobs



Skill intensity = skilled labor share at occupation level. Skilled = College graduate or higher. The size of circle is proportional to wage bill at the initial year.

Tool innovation Correlated with Wage Growth



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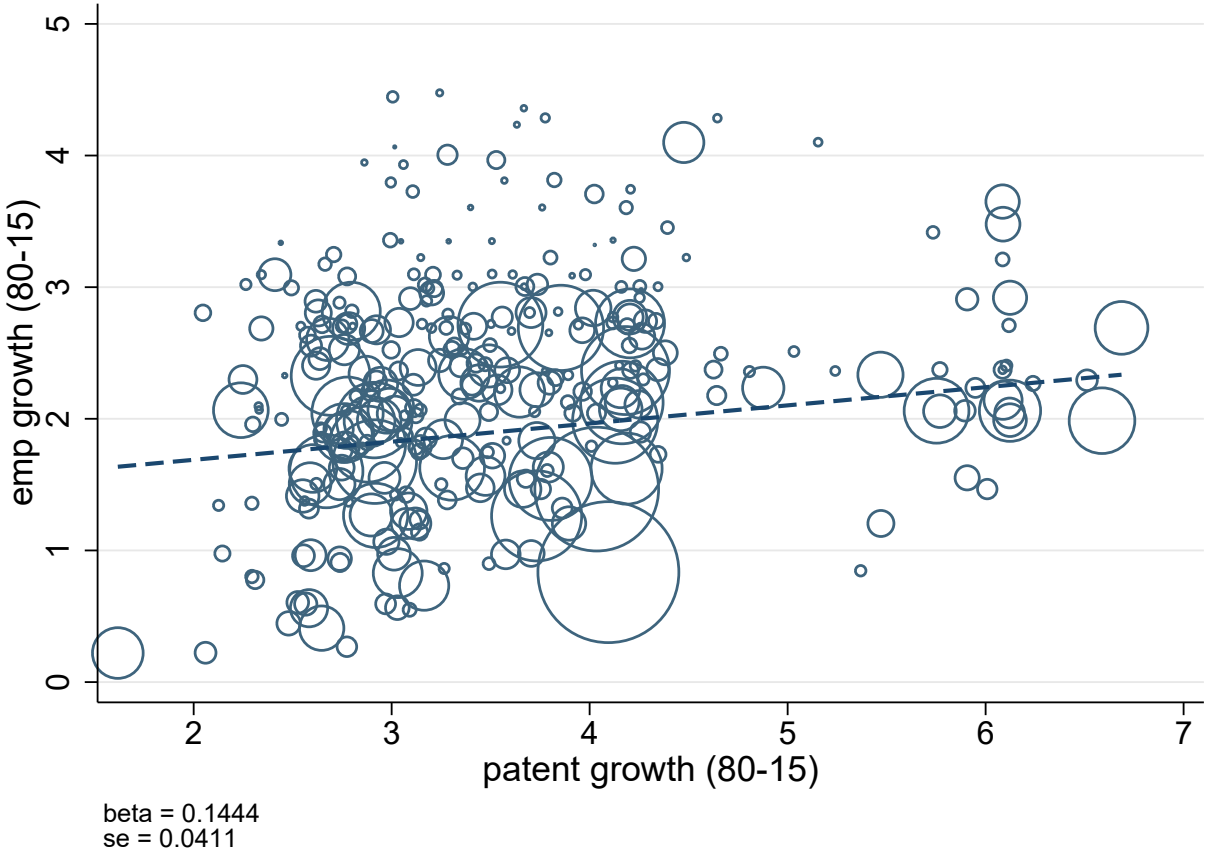
Tool innovation Correlated with Employment Growth



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Instrumental Variables for Innovation on Tools

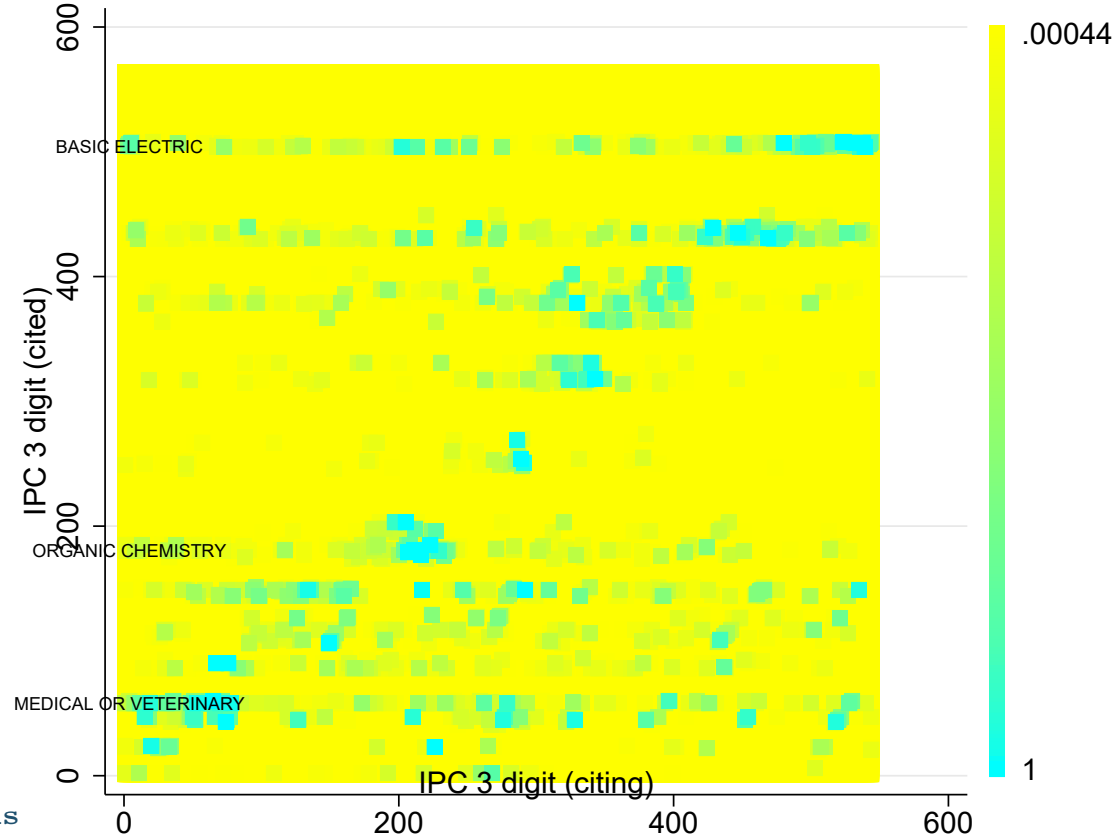
- Instrumental variable for tool innovation
- Concern : sectors with positive demand shock would have larger increase in innovation - endogeneity problem
  - e.g., innovation in computer sector $\rightleftarrows$ wage increase in IT sector
- Use technology sector that has knowledge spillover to the interested sector
  - Innovation in electricity is correlated with innovation in computer sector but not with demand shock for IT workers
  - Use the number of patents in "upstream" sector as an instrument
  - **Assumption:** When patent A cites B, knowledge spillover from B to A
  - Use patent citation data, construct the share of citations in patent class level, identify "upstream" sector
  - We exclude technology sectors that are in the same NAICS 2 digit

Construction of IV

- For patent class i,

$$\sum_{j \neq i} s_{ij} \Delta \log \#Patent_{jt} \quad \rightarrow \quad \Delta \log \#Patent_{it}$$

- s_{ij} is the citation share from technology class i to j
- The final IV is weighted with class shares of patents for o



IV Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent	0.130^{***}	0.0723	0.0417^{***}	0.0314^{***}
	(0.0156)	(0.0544)	(0.00992)	(0.00447)
N	324	324	323	324
F (first stage)	431.8	431.8	430.5	431.8

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period.

first stage with market value other specifications

★ How? Innovation reduces the price for tools. Innovation and Tool Price

Heterogeneous Occupations

- Different occupation may have different complementarity with tools
- Divide occupations according to their task: more vs less abstract
  - Occupational task measure from Autor and Dorn (2013)
  - Abstract task : creative, problem-solving, and coordination tasks
  - Occupations with highest abstract score : Chemical Engineers, Remote Sensing Scientists, Materials Engineers, Aerospace Engineers, Civil Engineers
  - Occupations with lowest abstract score : Packers and Packagers, Maintenance Workers, Material Collectors, Parking Lot Attendants

Task Scores across Occupation Bins

- No remarkable difference for manual and routine tasks scores
- Patent growth higher for abstract occupations

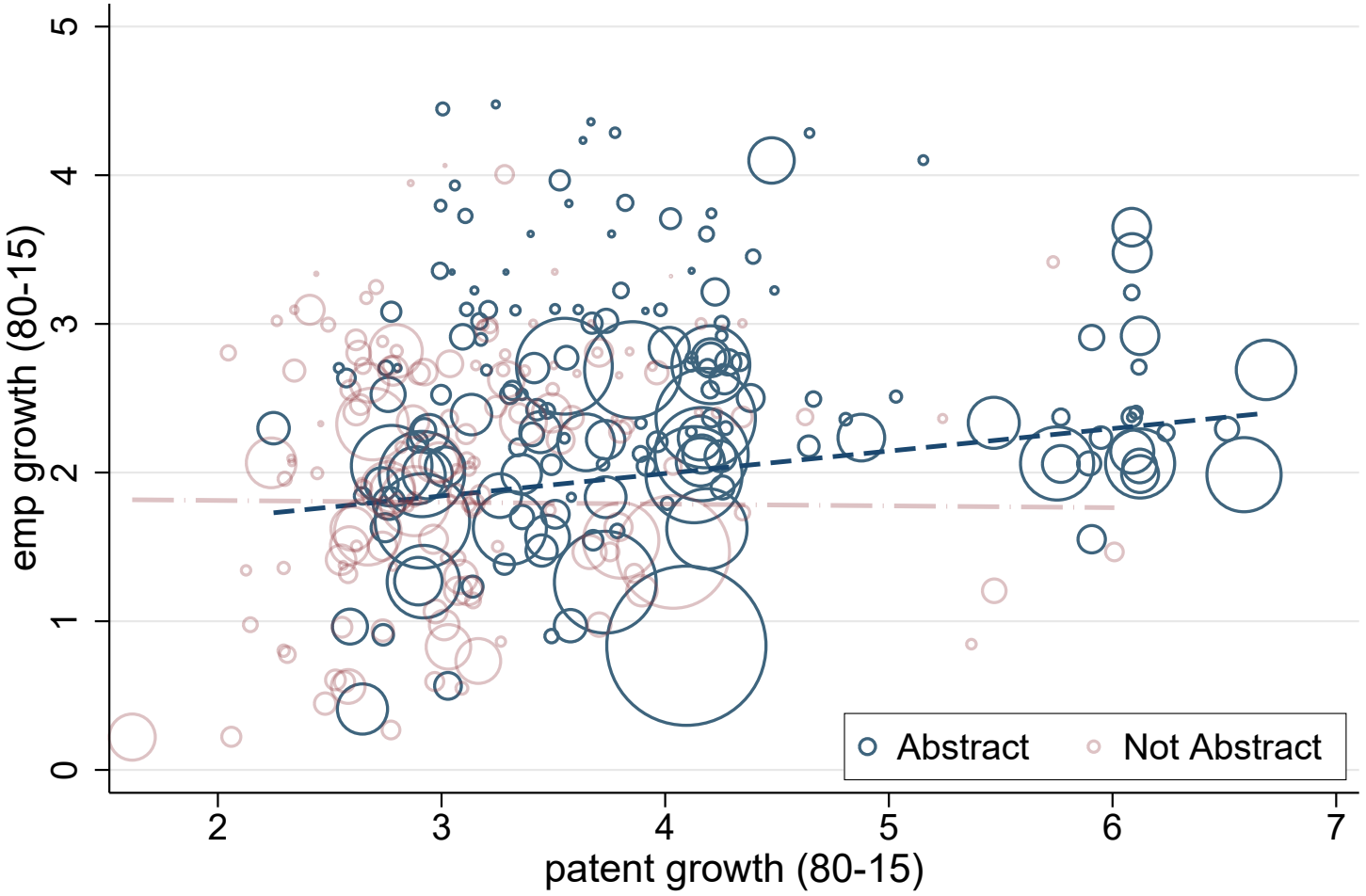
		Mean	Sd	Median	Min	Max
Routine Score	Not Abstract	4.50	2.21	4.95	1.19	8.64
	Abstract	4.01	2.24	3.41	1.25	8.34
Manual Score	Not Abstract	1.64	1.52	1.39	0.00	10.00
	Abstract	1.12	1.27	0.76	0.00	6.17
Abstract Score	Not Abstract	1.23	0.54	1.23	0.00	2.06
	Abstract	4.68	1.92	4.62	2.06	9.00
Patent growth (80-15)	Not Abstract	3.15	0.72	3.04	1.62	6.01
	Abstract	3.92	1.03	3.73	2.25	6.69

Labor Market Variables across Occupation Bins

- Abstract occupations have higher skill intensity

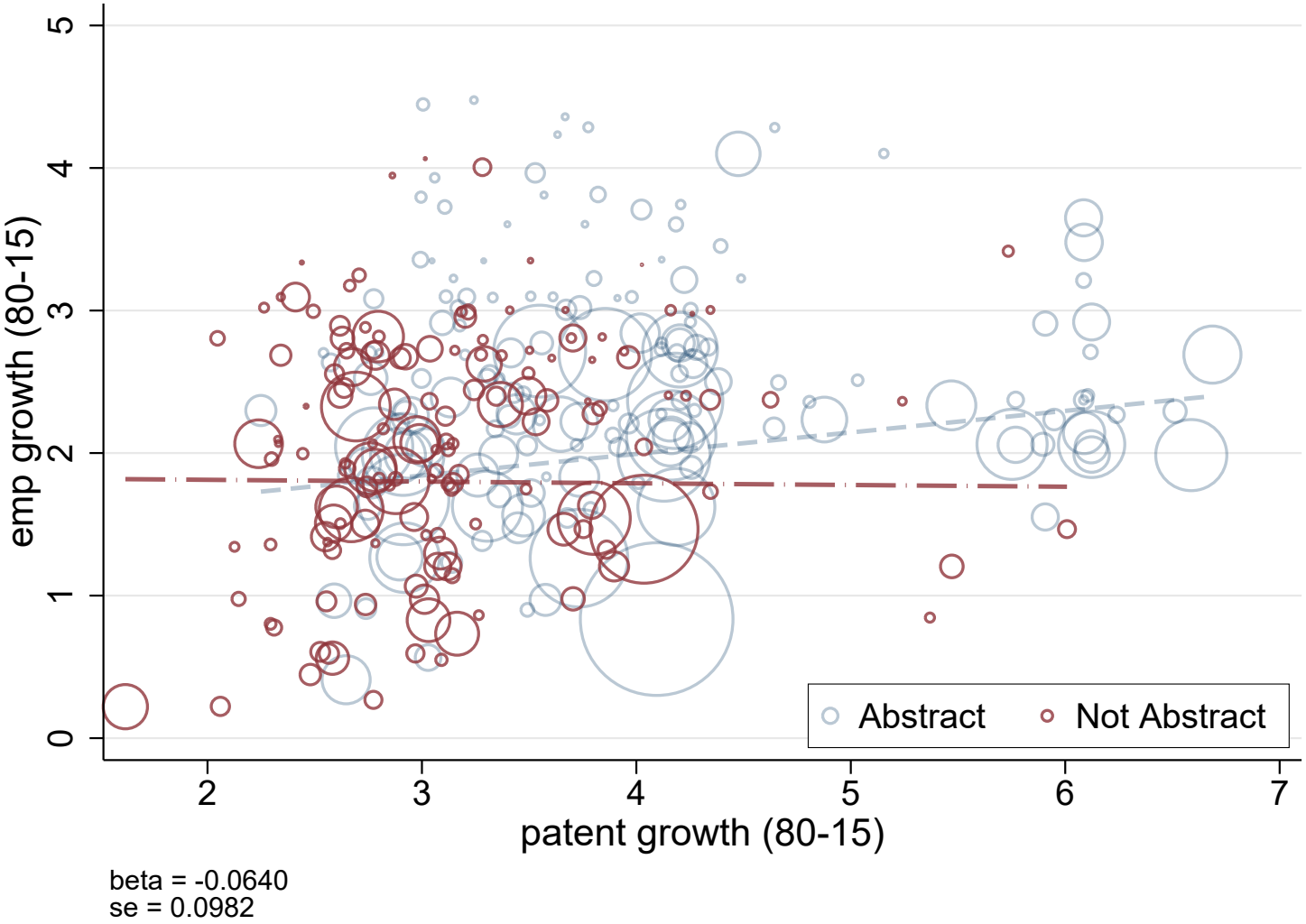
		Mean	Sd	Median	Min	Max
Skill Intensity (1980)	Not Abstract	0.27	0.12	0.30	0.04	0.63
	Abstract	0.47	0.18	0.39	0.18	0.98
Skill Premium (1980)	Not Abstract	1.12	0.15	1.09	0.74	1.71
	Abstract	1.26	0.21	1.22	0.89	2.06
Wage Growth (80-15)	Not Abstract	1.19	0.21	1.16	0.82	1.88
	Abstract	1.36	0.25	1.35	0.82	2.76
Emp. Growth (80-15)	Non Abstract	1.96	1.01	2.03	-1.84	5.70
	Abstract	2.60	0.86	2.49	0.41	5.14

Patent and Employment across Occupation Bins

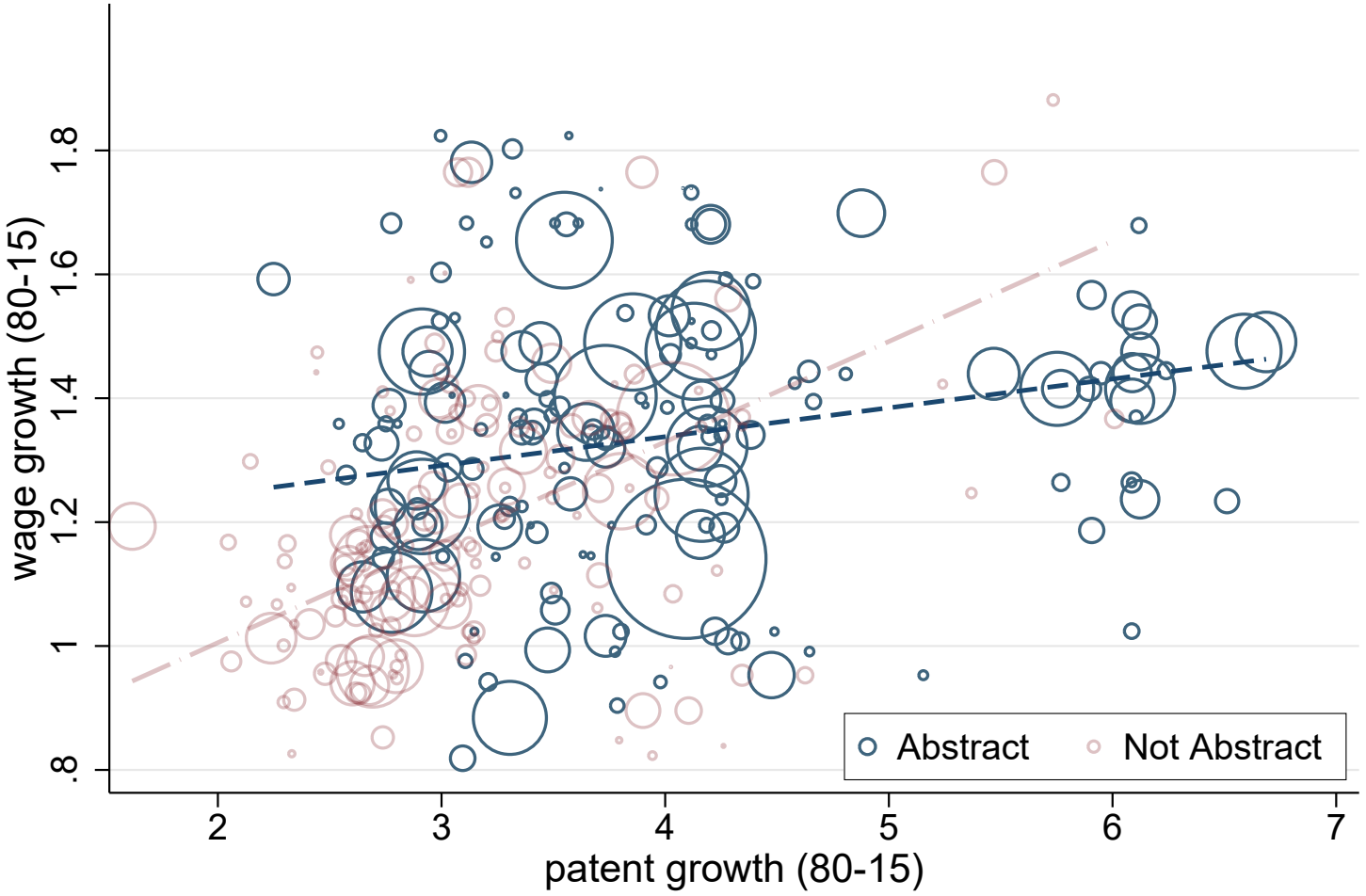


beta = 0.1518
se = 0.0510

Patent and Employment across Occupation Bins

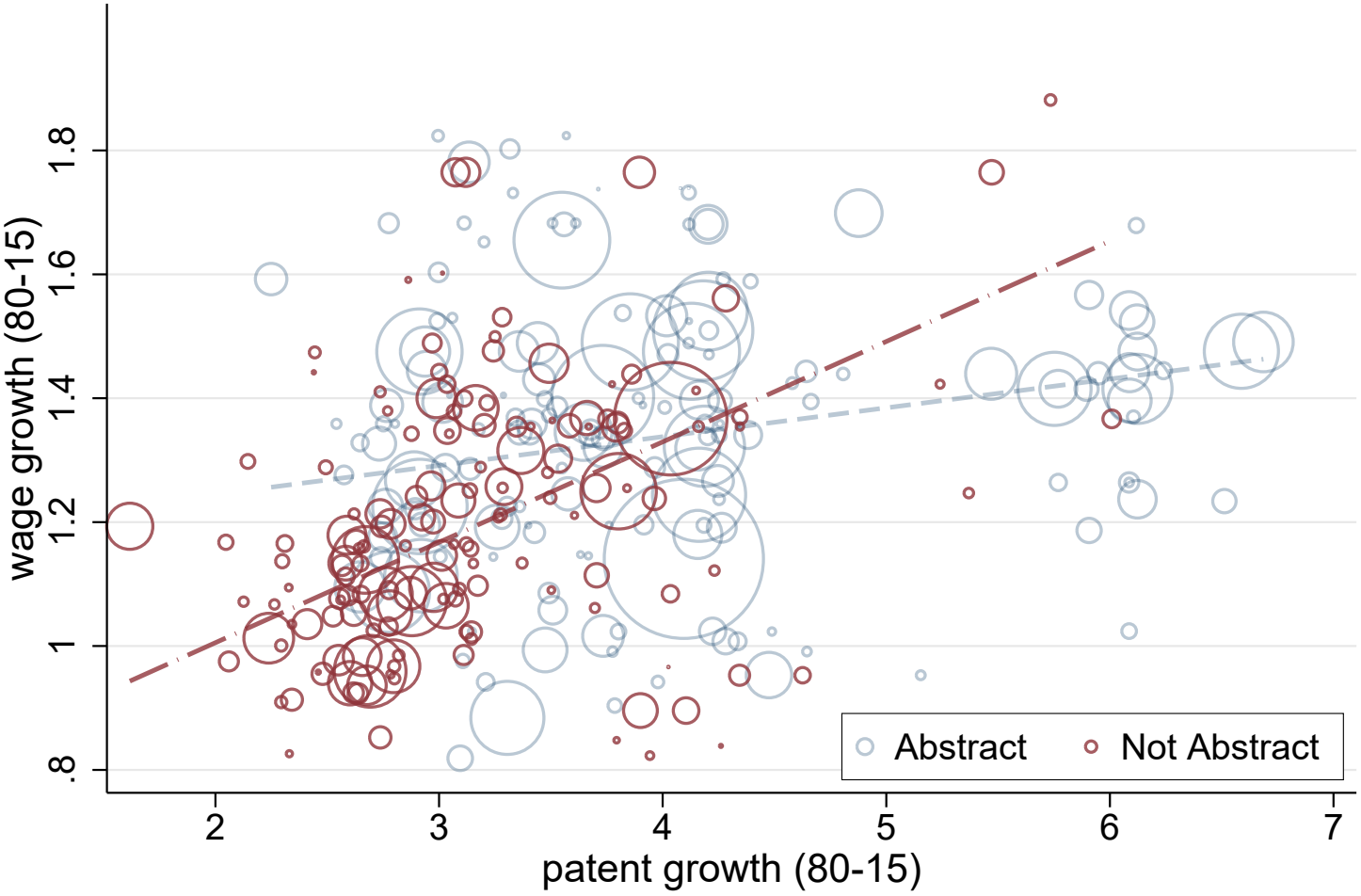


Patent and Wage across Occupation Bins



beta = 0.0359
se = 0.0163

Patent and Wage across Occupation Bins



beta = 0.1621
se = 0.0197

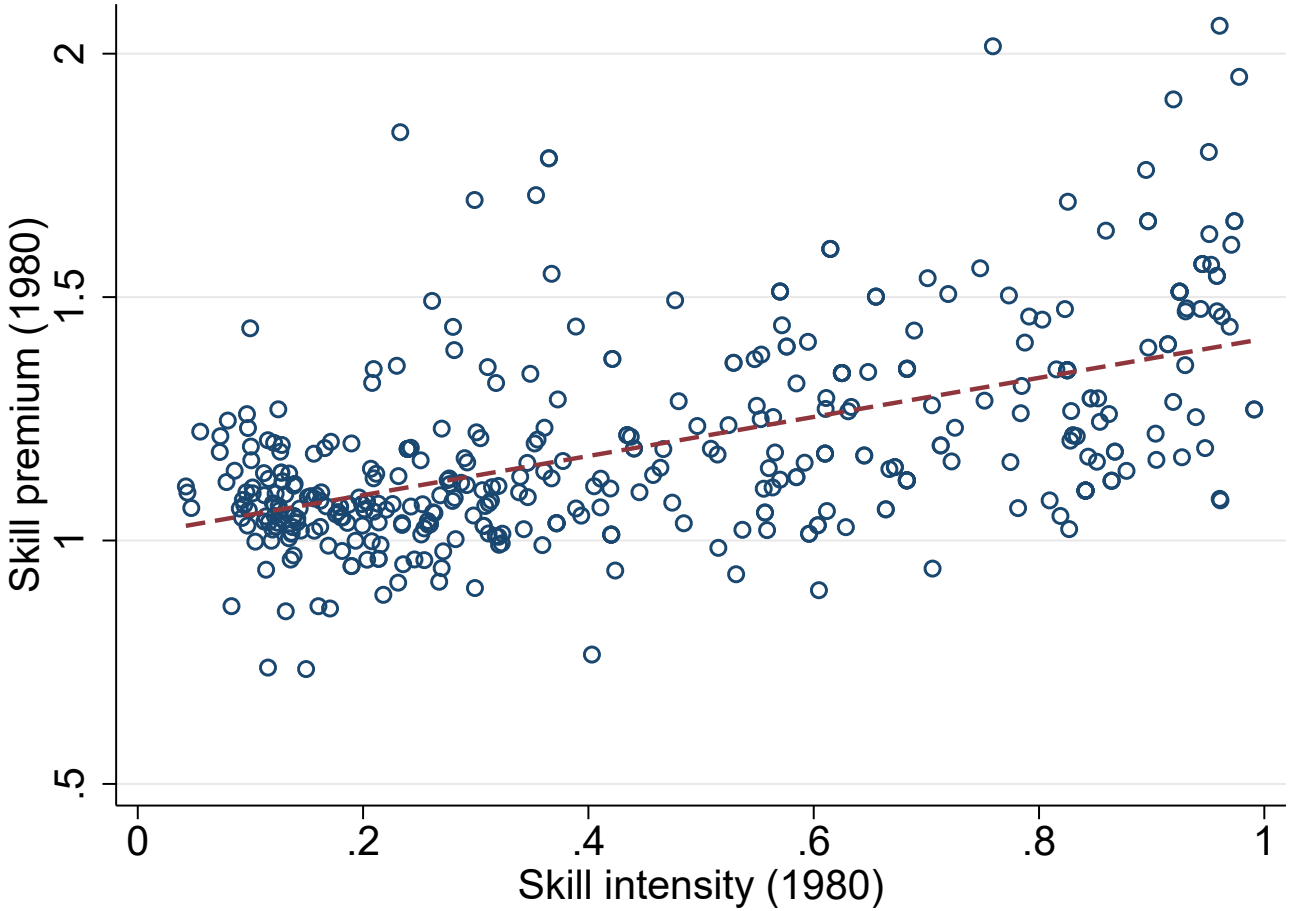
Conclusion

- Novel way to measure technology change at the occupation level
- New findings about the tool-embodied technological change
  - Technological change is more directed toward skill-intensive jobs
  - Technological change is overall labor-augmenting
  - Technological change is less labor-augmenting for non-abstract jobs
- Currently making distinction between task-substituting vs. task-enhancing tools
  - Task-substituting if occupational tasks = tool function
    e.g. typewriter vs. computer
  - Task-enhancing if tool function complements occupational tasks
    e.g. engineers use computer

$\Rightarrow$ (Expectation) Tools are more task-enhancing for abstract jobs.

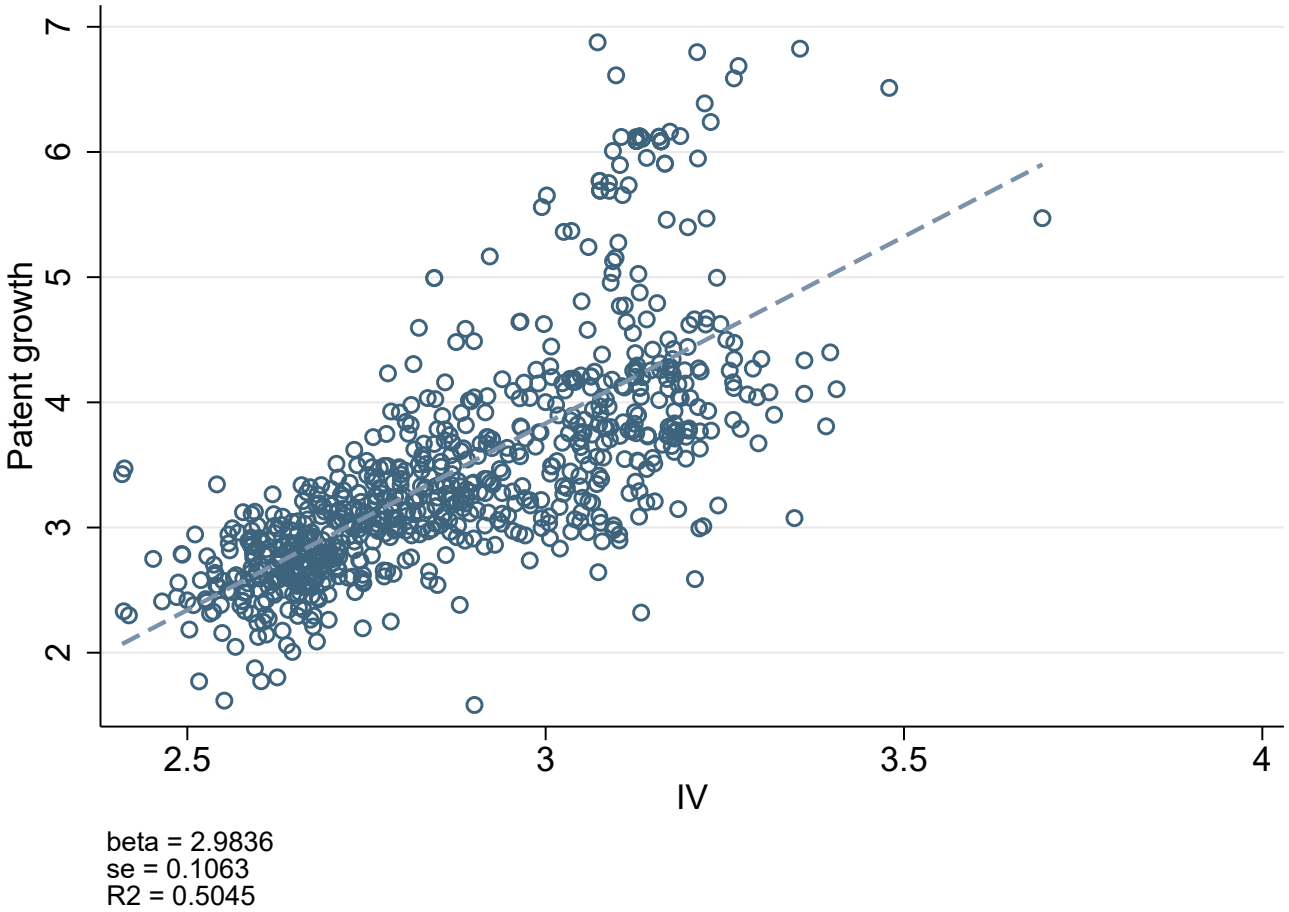
Appendix

Skill Premium and Skill Intensity



Skill premium = Average wage of skilled / Average wage of unskilled at occupation level. Skill intensity = share of skilled workers. Skilled = College graduate or higher back

Skill Premium and Skill Intensity



IV $= \sum_i s_{io} \sum_{j \neq i} s_{ij} \Delta \log P_{jt}$, where s_{ij} is patent class j's share in patent class i's references, and s_{io} is the share of patent class i in tools in occupation o. back

Example of tool innovation at occupation level

Occupation	Tools	Wikipedia found	Patents
Bakers	Automatic dough presses	Yes	15
Bakers	Bagel dividers	Yes	10
Bakers	Baking frames	Yes	10
Bakers	Baking sheets	Yes	5
Bakers	Balance scales	Yes	5
Bakers	Barquette molds	No	.

Then the tool innovation of bakers is $(15+10+10+5+5)/5 = 9$ back

Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 p_{i,1980-2015} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.334^{***}	0.200^{***}	0.121	0.174^{***}	0.0711^{**}	0.448^{***}
	(0.0663)	(0.0502)	(0.156)	(0.00679)	(0.0244)	(0.0364)
N	326	325	326	20467	14733	20616
F (first stage)	50.76	50.57	50.76	3580.1	2621.7	3227.6
sector fixed	no	no	no	yes	yes	yes

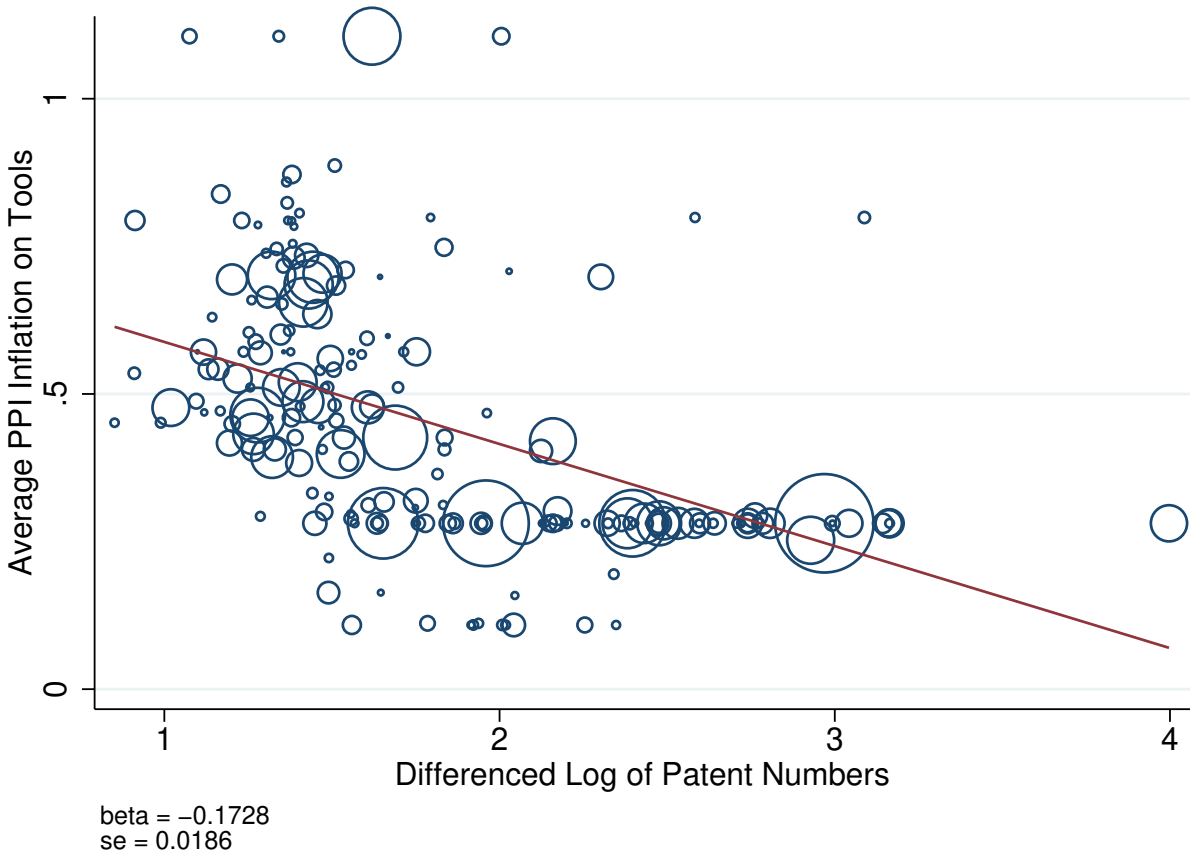
Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit

back

More Innovation Reduces Price



back

Effects of Cheaper Tools on Labor Market - OLS

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 ppigr_{i,1980-2015} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Wage	Col. Pre.	Emp.	Skill Int.	Wage	Col. Pre	Emp.	Skill Int.
PPI	-0.0621	-0.0702	-0.339	-0.111***	-0.194***	-0.0122	-0.252***	-0.00822*
	(0.126)	(0.0478)	(0.230)	(0.0205)	(0.0101)	(0.0334)	(0.0412)	(0.00406)
N	145	145	145	145	13288	9660	13387	13387
sector fixed	no	no	no	no	yes	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Effects of Cheaper Tools on Labor Market - IV

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 ppigr_{i,1980-2015} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Wage	Col. Pre.	Emp.	Skill Int.	Wage	Col. Pre	Emp.	Skill Int.
PPI	-0.266	-0.361***	0.246	-0.181***	-0.533***	-0.313***	0.0680	-0.0208*
	(0.213)	(0.0900)	(0.395)	(0.0358)	(0.0214)	(0.0679)	(0.0836)	(0.00824)
N	145	145	145	145	13288	9660	13387	13387
F (first stage)	77.00	77.00	77.00	77.00	4208.6	3054.8	4241.0	4241.0
sector fixed	no	no	no	no	yes	yes	yes	yes

Standard errors in parentheses

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back

GMM Estimation - Details

$$\mathbb{E}\left[\left\{\Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right)\right]\right\}\Delta_t \log z_{ot}^i\right]$$

$$= \mathbb{E}\left[\Delta_t \log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\lambda_{ot}}{1-\lambda_{ot}}\right)\right]\right]\mathbb{E}\left[\Delta_t \log z_{ot}^i\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{w_{ot}^s}{w_{ot}^u}\right) - \Delta_t \alpha_t - \frac{\sigma-1}{\sigma}(1-\delta)\Delta_t \log(x_{ot}) + \frac{1}{\sigma}\Delta_t \log\left(\frac{s_{ot}}{u_{ot}}\right)\right)\left(\Delta_t \log z_{ot}^i\right) \approx 0$$

$$\mathbb{E}\left[\left\{\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right)\right]\right\}\Delta_t \log z_{ot}^i\right]$$

$$= \mathbb{E}\left[\Delta_t \log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right) - \Delta_t \mathbb{E}_t\left[\log\left(\frac{\alpha_{ot}}{\alpha_{1t}}\right)\right]\right]\mathbb{E}\left[\Delta_t \log z_{ot}^i\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{w_{ot}^s}{w_{1t}^s}\right) - \Delta_t \beta_t - \frac{\rho-\sigma}{\rho(\sigma-1)}\Delta_t \log\left(\frac{\theta_{ot}}{\theta_{1t}}\right) - \frac{\sigma-1}{\sigma}\Delta_t \log\left(\frac{x_{ot}}{x_{1t}}\right)\right.$$

$$\left. - \frac{\rho-\sigma}{\rho\sigma}\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right)\right)\left(\Delta_t \log z_{ot}^i\right) \approx 0$$

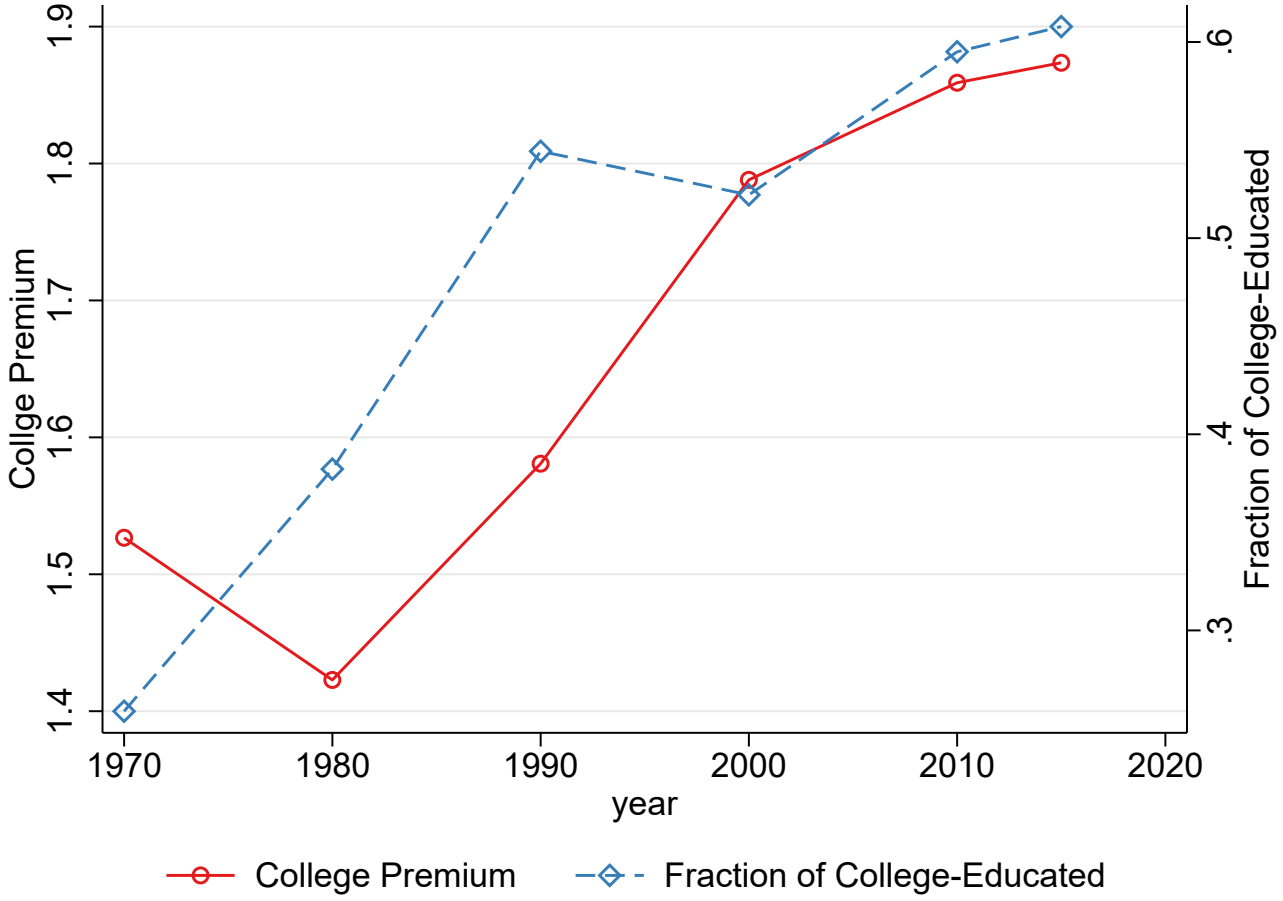
$$\mathbb{E}\left[\eta\left(\Delta_t \epsilon_{ot}^s - \Delta_t \epsilon_{1t}^s - \Delta_t \mathbb{E}_t\left[\epsilon_{ot}^s - \epsilon_{1t}^s\right]\right)\Delta_t x_{ot}\right] = \eta \mathbb{E}\left[\Delta_t \epsilon_{ot}^s - \Delta_t \epsilon_{1t}^s - \Delta_t \mathbb{E}_t\left[\epsilon_{ot}^s - \epsilon_{1t}^s\right]\right]$$

$$\times\, \mathbb{E}\left[\Delta_t \log x_{ot}\right] = 0$$

$$\Rightarrow \frac{1}{N_o}\sum_o \left(\Delta_t \log\left(\frac{s_{ot}}{s_{1t}}\right) - \Delta_t \gamma_t - \eta \Delta_t \log\left(\frac{w_{ot}^s}{w_{1t}^s}\right)\right)\left(\Delta_t \log x_{ot}\right) \approx 0$$

back

Rising Skill Premium and Supply of Skilled Workers



Skill Premium = Mean wage of college or above/Mean wage of high school or below

back

Rising Skill Premium: Why?

- The literature has emphasized **skill-biased technical change**.
  - Define the aggregate production function as

$$Y = \left[(A_l L)^{(\sigma-1)/\sigma} + (A_h H)^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}$$

  - Assuming competitive wages, skill premium is simply

$$\omega = (A_h/A_l)^{(\sigma-1)/\sigma} (H/L)^{-1/\sigma}$$

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$$\omega = (A_h/A_l)^{(\sigma-1)/\sigma}(H/L)^{-1/\sigma}$$

Table: Implied A_h/A_l, from Acemoglu (2002)

	Some college	College graduate
1950	0.146	0.219
1960	0.189	0.343
1970	0.199	0.485
1980	0.270	0.643
1980	0.357	1.064

Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent	0.0845^{***}	0.144^{***}	0.0440^{***}	0.0165^{***}
	(0.0115)	(0.0411)	(0.00753)	(0.00329)
N	324	324	323	324

Standard errors in parentheses

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Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. other specification back

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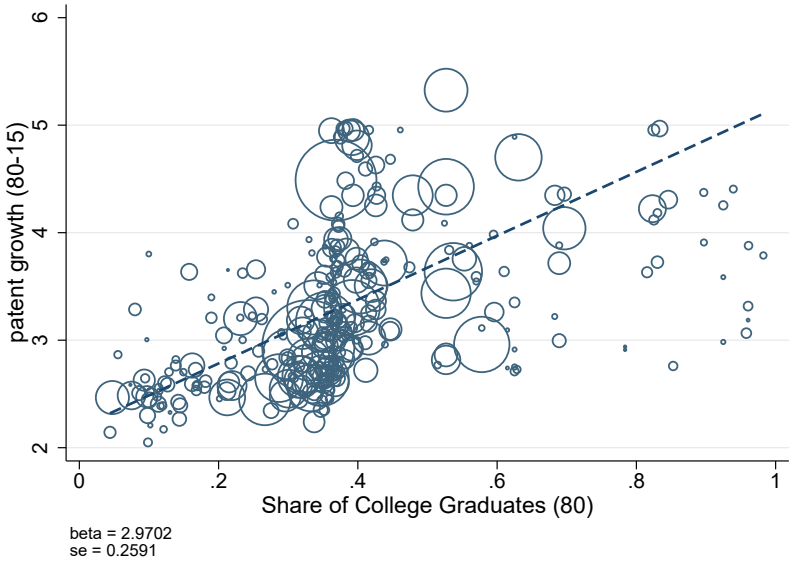
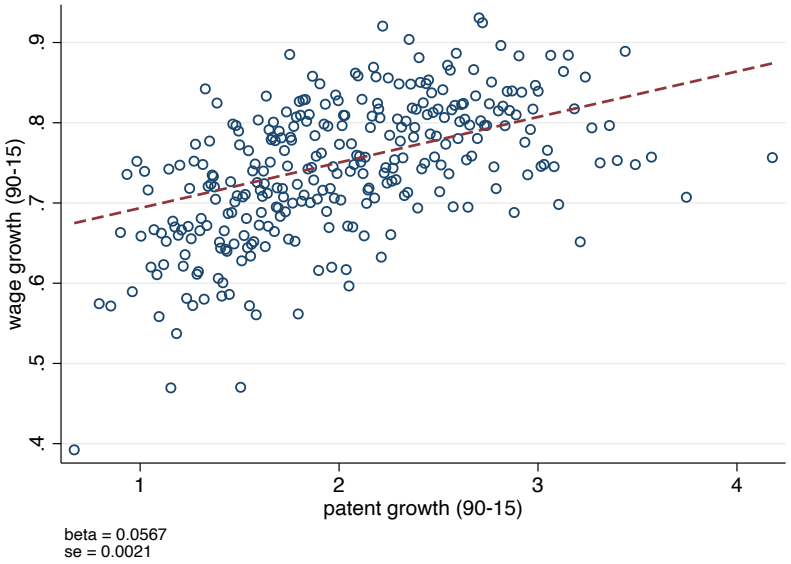
Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit first stage other specification

★ How? Innovation reduces the price for tools. Innovation and Tool Price

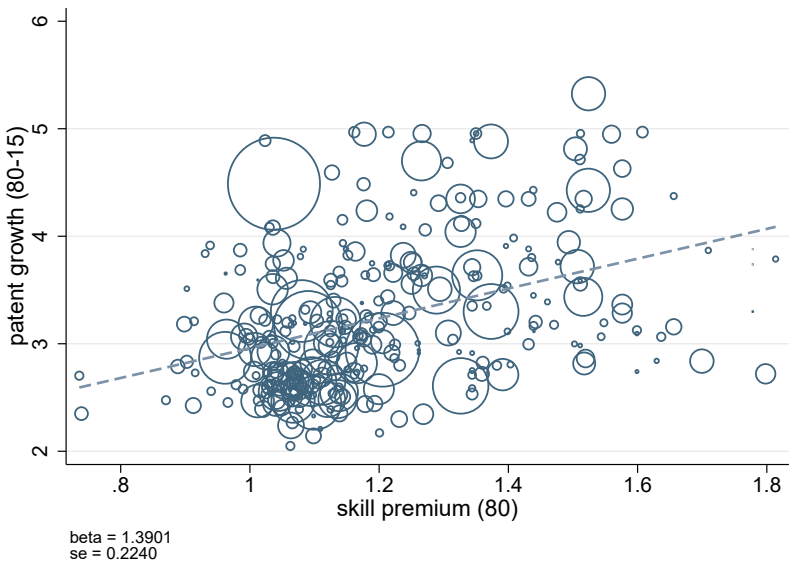
Skill-biased Innovation and Wage Increase

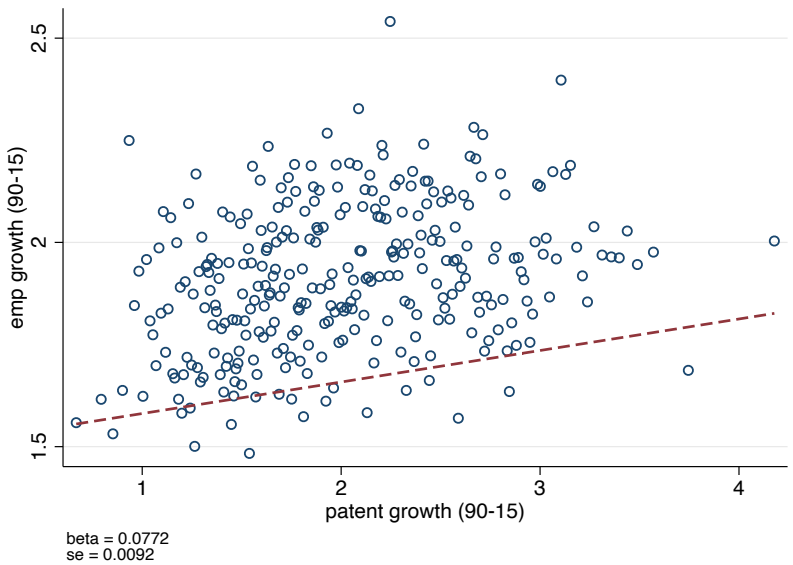




Skill intensity = skilled labor share at occupation level. Skilled = College graduate or higher. Wage growth is residual after controlling industry fixed effects. back

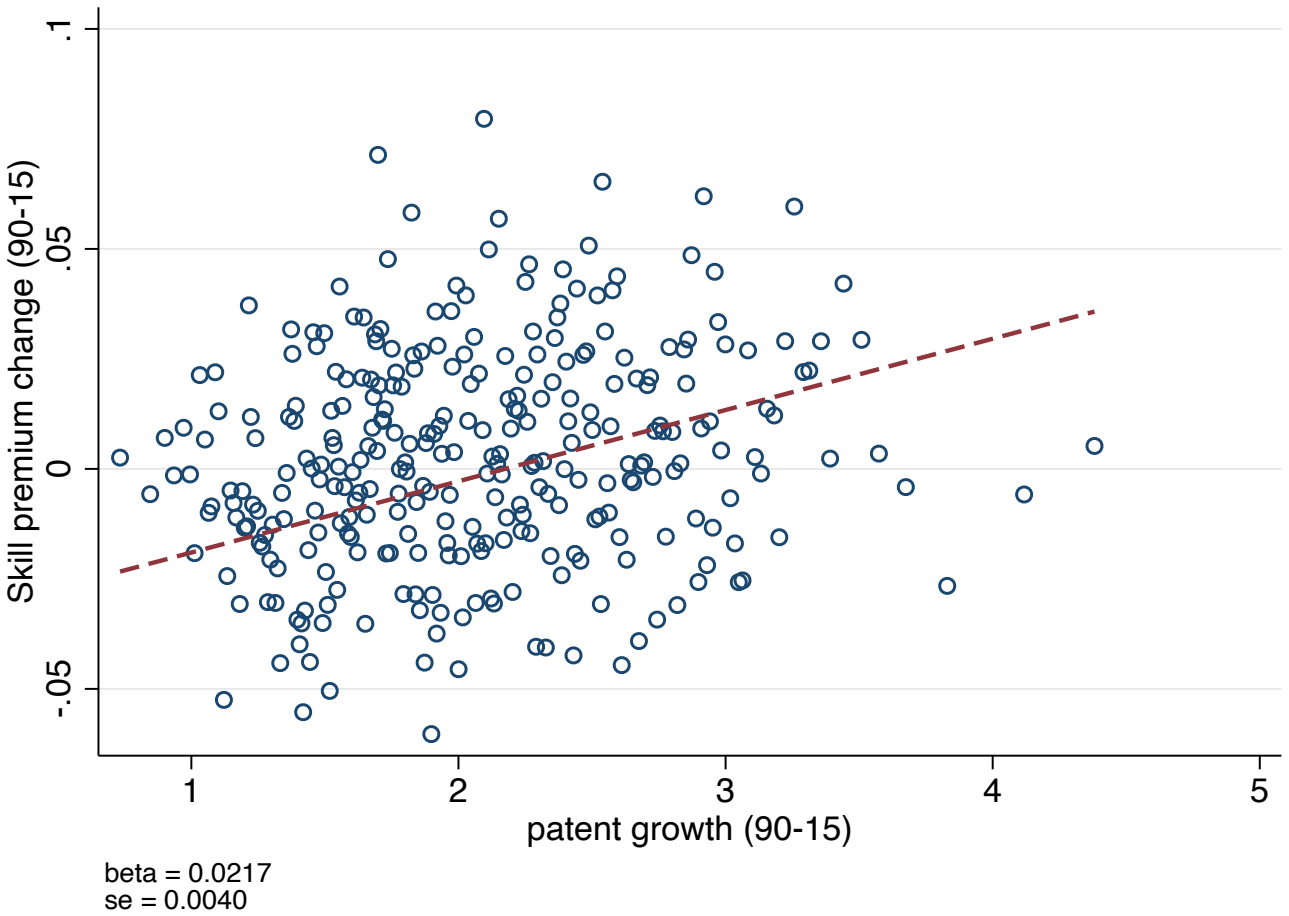
Skill-biased Innovation and Employment Increase





Skill premium = Average wage of skilled / Average wage of unskilled at occupation level. Skilled = College graduate or higher. Emp growth is residual after controlling industry fixed effects. back

Tool Innovation and Skill Premium Increase



Skill premium = Average wage of skilled / Average wage of unskilled at occupation level.
Skilled = College graduate or higher. Residuals after controlling industry fixed effects.

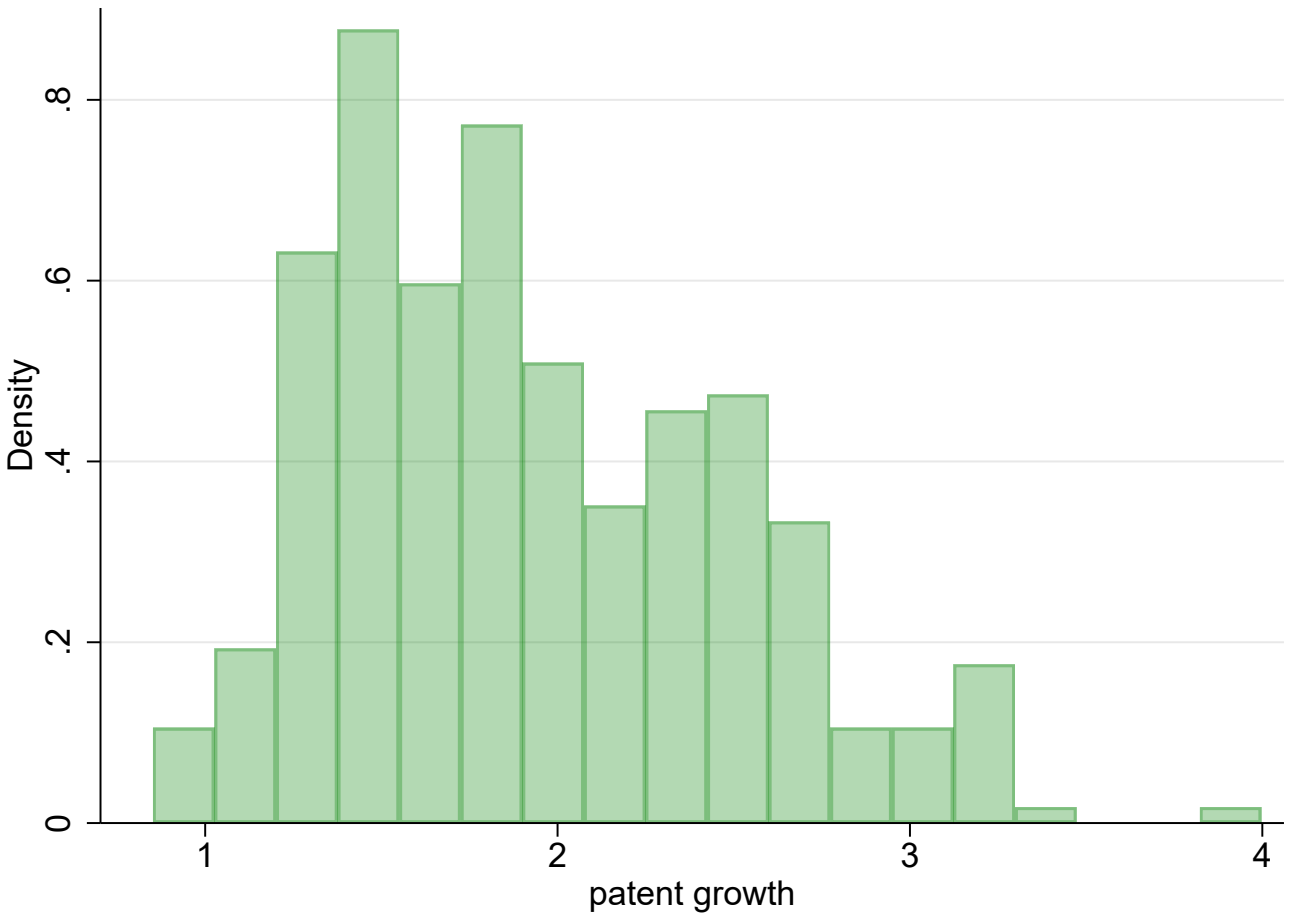
back

Compare with Webb's JMP

- This project
  - Measure innovation on **Tools** → Occupations
  - Study all tools provided by ONET
  - Assuming tools are **complements** to labor
- Webb's JMP
  - Measure technology on **Tasks** of occupations
  - Case study : AI, software, and robots
  - Assuming AI and robots are **substitutes** to labor at task level

back

Distribution of Patent growth at occupation level



back

Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \log p_{i,1980-2015} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.0182	0.0267	-0.126	0.00207	0.00445	0.0947***
	(0.0171)	(0.0158)	(0.0865)	(0.00234)	(0.00954)	(0.0174)
N	326	325	326	20467	14733	20616
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \log p_{i,1980-2015} + \beta_2 \log p_{i,1970-1980} + \epsilon_i$$

	(1)	(2)	(3)	(4)	(5)	(6)
	Wage	Col. Pre.	Emp.	Wage	Col. Pre	Emp.
log(patent)	0.0644***	0.0558***	-0.0426	0.0356***	0.0114	0.169***
	(0.0171)	(0.0165)	(0.0956)	(0.00253)	(0.0105)	(0.0194)
N	326	325	326	20467	14733	20616
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

Regression Result - using market value of patents

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent (value)	0.0458^{***}	0.0898^{**}	0.0316^{***}	0.0112^{***}
	(0.0103)	(0.0353)	(0.00650)	(0.00283)
N	324	324	323	324

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Regression is weighted by wage bill in the initial period. Market value of patent is from Kogan et al.(2017). back

Regression Result - using citation weighted patents

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent (cite)	0.0854^{***}	0.138^{***}	0.0440^{***}	0.0170^{***}
	(0.0112)	(0.0403)	(0.00735)	(0.00320)
N	324	324	323	324

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Regression is weighted by wage bill in the initial period. We use residual of citations after regressing with application year. back

Regression Result - with sector fixed effects

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1980-2015} + \delta_{s(i)} + \epsilon_i$$

	Wage	Emp.	Skill Pre	Skill Int.
patent	0.0300***	0.152***	0.00343	-0.00648***
	(0.00131)	(0.00704)	(0.00490)	(0.000710)
N	18,554	18,638	14,539	19,544
sector fixed	yes	yes	yes	yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

IV Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 p_{i,1980-2015} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.334^{***}	0.200^{***}	0.121	0.174^{***}	0.0711^{**}	0.448^{***}
	(0.0663)	(0.0502)	(0.156)	(0.00679)	(0.0244)	(0.0364)
N	326	325	326	20467	14733	20616
F (first stage)	50.76	50.57	50.76	3580.1	2621.7	3227.6
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

IV Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \log p_{i,1980-2015} + \beta_2 \log p_{i,1970-1980} + \epsilon_i$$

	(1) Wage	(2) Col. Pre.	(3) Emp.	(4) Wage	(5) Col. Pre	(6) Emp.
log(patent)	0.114^{***}	0.0708^{**}	0.0570	0.0608^{***}	0.0371^{**}	0.202^{***}
	(0.0228)	(0.0217)	(0.117)	(0.00335)	(0.0139)	(0.0234)
N	326	325	326	20467	14733	20616
F (first stage)	435.6	434.9	643.5	26698.8	19200.0	43609.6
sector fixed	no	no	no	yes	yes	yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

IV Regression Result - market value weighted

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent (value)	0.137^{***}	0.111^{**}	0.0407^{***}	0.0299^{***}
	(0.0176)	(0.0541)	(0.00999)	(0.00463)
N	324	324	323	324
F (first stage)	235.6	235.6	234.9	235.6
sector fixed	no	no	no	no

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. Sector is NAICS 3 digit. Market value of patent is from Kogan et al.(2017).

back

IV Regression Result - citation weighted

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent (cite)	0.126***	0.0725	0.0399***	0.0303***
	(0.0144)	(0.0509)	(0.00926)	(0.00414)
N	324	324	323	324
F (first stage)	543.0	543.0	541.3	543.0
sector fixed	no	no	no	no

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Regression is weighted by wage bill in the initial period. We use residual of citations after regressing with application year. back

IV Regression Result - with sector fixed effects

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta \log p_{i,1980-2015} + \delta_{s(i)} + \epsilon_i$$

	Wage	Emp.	Skill Pre	Skill Int.
patent	0.0411***	-0.0397***	0.0167**	0.00492***
	(0.00184)	(0.0100)	(0.00685)	(0.000999)
N	18554	18638	14539	19544
F (first stage)	19125.6	19212.9	14932.3	20150.3
sector fixed	yes	yes	yes	yes

Standard errors in parentheses

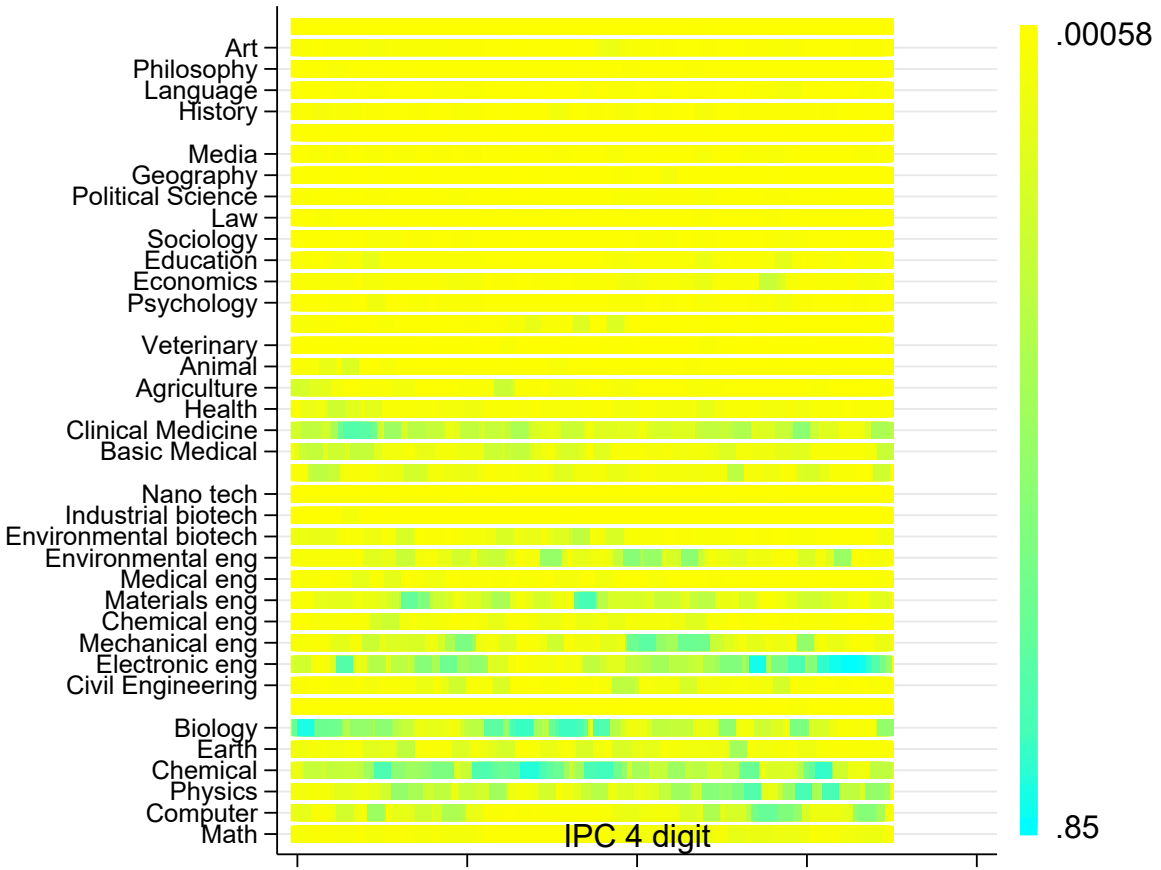
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Regression is weighted by wage bill in the initial period. Sector is NAICS 3 digit. back

Construction of IV

$$\sum_{j \neq i} s_{ij} \Delta \log P_{jt} \quad \rightarrow \quad \Delta \log P_{it}$$

- Use science field that has knowledge spillover to the interested sector
- s_{ij} is the citation share from technology class i to Science field j



IV Regression Result

$$\Delta y_{i,1980-2015} = \beta_0 + \beta_1 \Delta p_{i,1980-2015} + \epsilon_i$$

	Wage	Emp.	Skill Pre.	Skill Int.
patent	0.0897^{***}	0.0696	0.0446^{***}	0.0172^{***}
	(0.0160)	(0.0573)	(0.0104)	(0.00456)
N	324	324	323	324
F (first stage)	344.5	344.5	343.4	344.5
sector fixed	no	no	no	no

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Wage and employment is logged. Wage and college premium is weighted by wage bill in the initial period. back